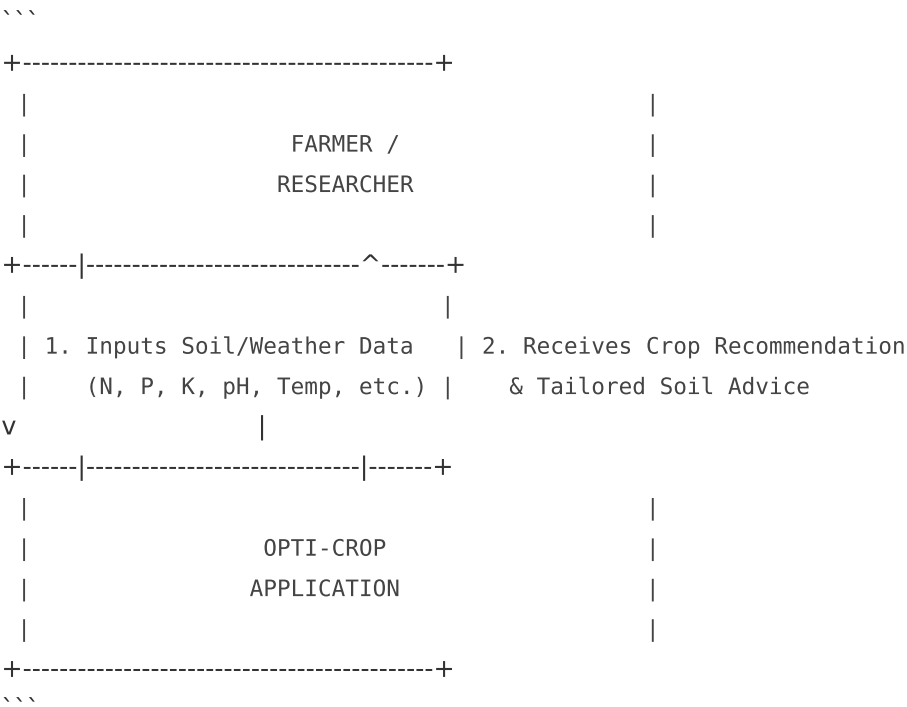


Data Flow Diagram (DFD)

1. Level 0 DFD (System Context Diagram)

The Context Diagram represents the boundaries of the Opti-Crop system, showing inputs from users and outputs generated by the application.



2. Level 1 DFD (Process Level Diagram)

The Level 1 DFD details how data flows between UI processes, the Machine Learning inference engine, and the SQLite storage database.

1. **Process 1.0: User Management**
- User inputs credentials (Register/Login).
 - Verifies credentials against the `users` table in `opticrop.db`.
 - Establishes user session.
2. **Process 2.0: ML Prediction Engine**
- Receives soil features vector (1x7 array).
 - Loads trained model (`model.pkl`).

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- Executes prediction -> outputs crop label.

3. **Process 3.0: Advice & Analytics Compiler**

- Compares user inputs with optimal parameters loaded from `crop\_ranges.pkl`.
- Queries `crops` database table for water and seasonal details.
- Computes nutrient deviations and drafts chemical/organic remediation advice.

4. **Process 4.0: Log Storage & History**

- Saves inputs to `soil\_data` table.
- Saves prediction to `predictions` table.
- Saves summary to `reports` table.
- Updates user dashboard history.